Quantitative Methods I

Week 5 - Properties of a Function

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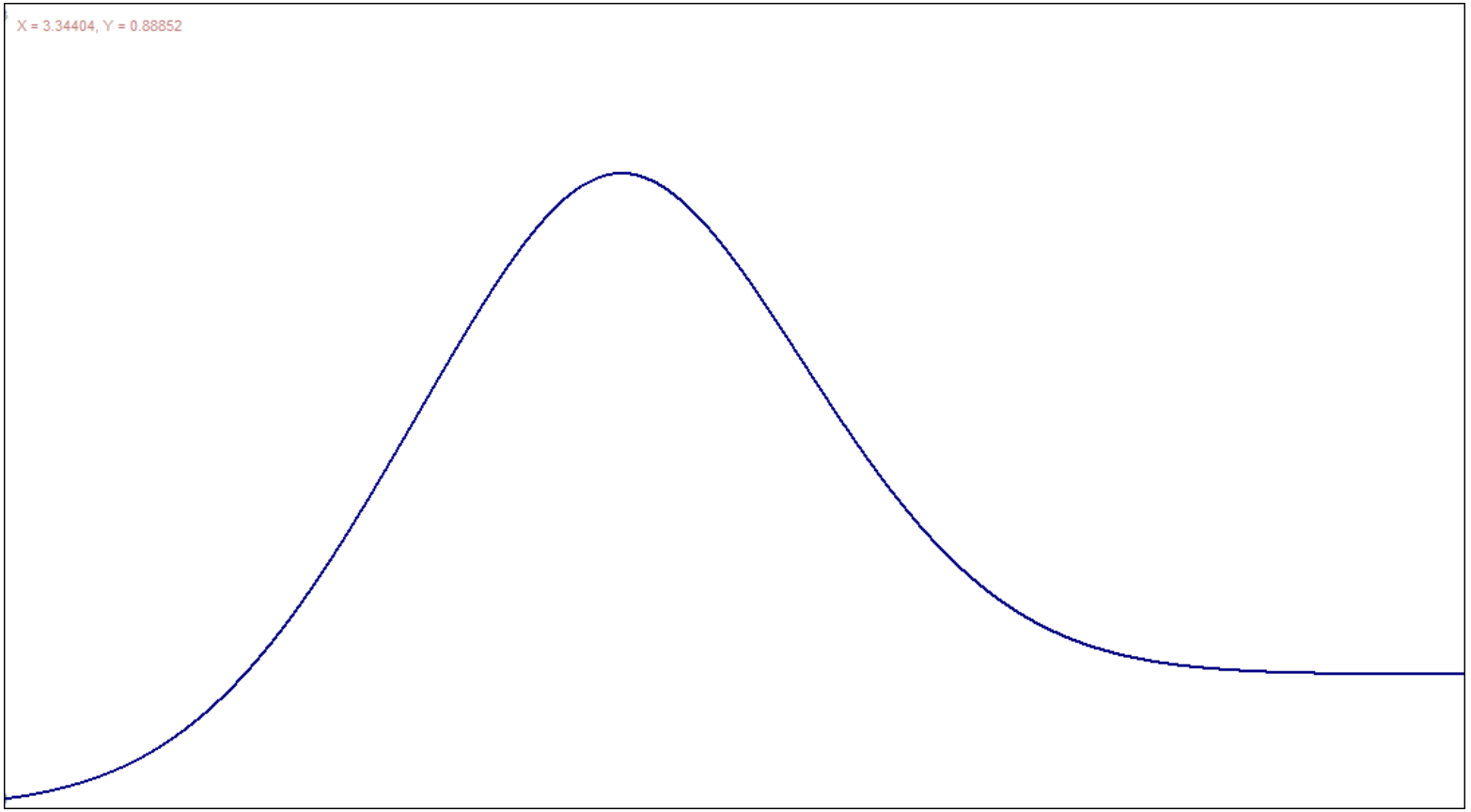
Motivation



- What properties of the depicted (profit) function may be important?
- How do these properties relate to the derivative?

Increasing/Decreasing

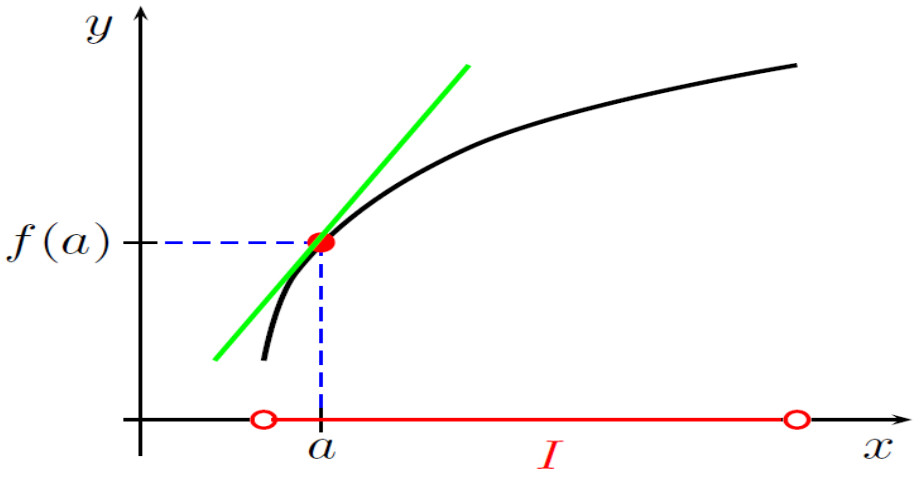
Conditions for now:

- the domain is restricted to a single open interval
  - this excludes the endpoints!
  - examples: $(1, 5)$, $(-1, \infty)$, $(-\infty, \infty)$
- the function is differentiable on this interval

Then,

- The function f is (strictly) increasing on interval I if, for any $x_1, x_2 \in I$ with $x_1 < x_2$, we have $f(x_1) < f(x_2)$
- The function f is (strictly) decreasing on interval I if, for any $x_1, x_2 \in I$ with $x_1 < x_2$, we have $f(x_1) > f(x_2)$

Increasing/Decreasing

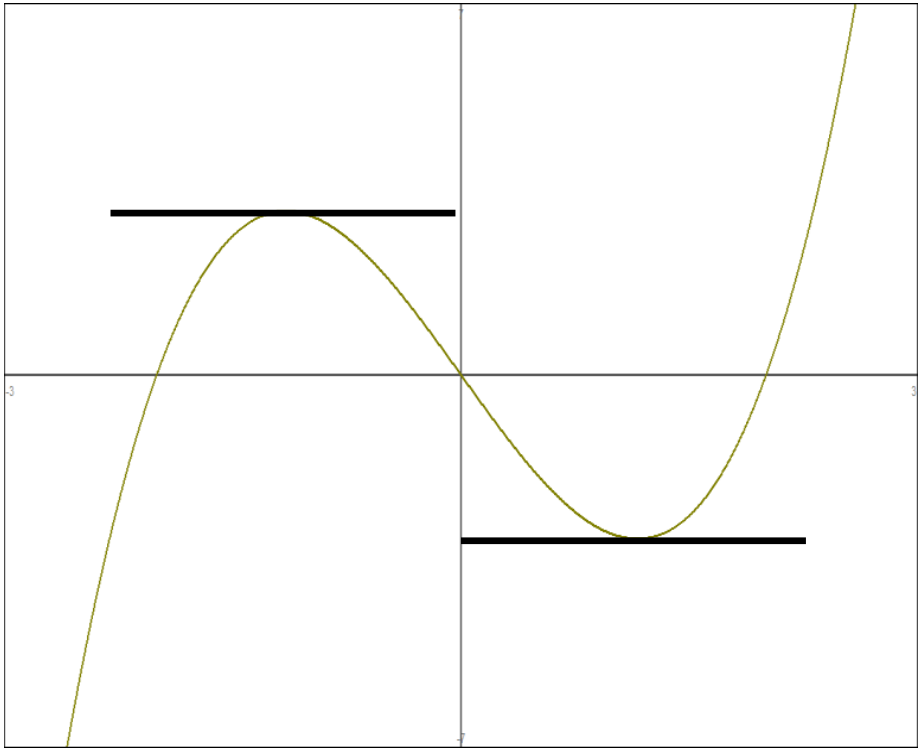


f increasing at a $\Leftrightarrow$ Tangent to f at a increasing

Therefore,

- If $f'(x) > 0$, for all x in the domain I of f, then f is (strictly) increasing on I
- If $f'(x) < 0$, for all x in the domain I of f, then f is (strictly) decreasing on I

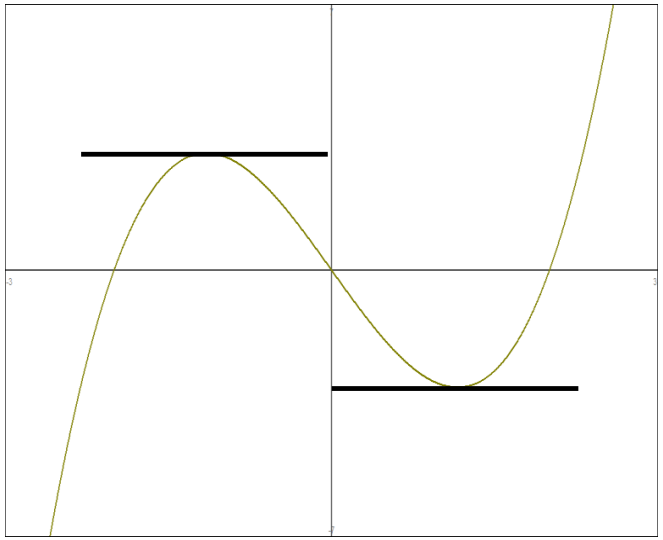
Example



Depicted is the function $f(x) = x^3 - 3x$. Since $f'(x) = 3x^2 - 3$, have

$$f'(x) = 0 \quad \Leftrightarrow \quad x^2 = 1 \quad \Leftrightarrow \quad x = 1 \text{ or } x = -1$$

Example - cont'd



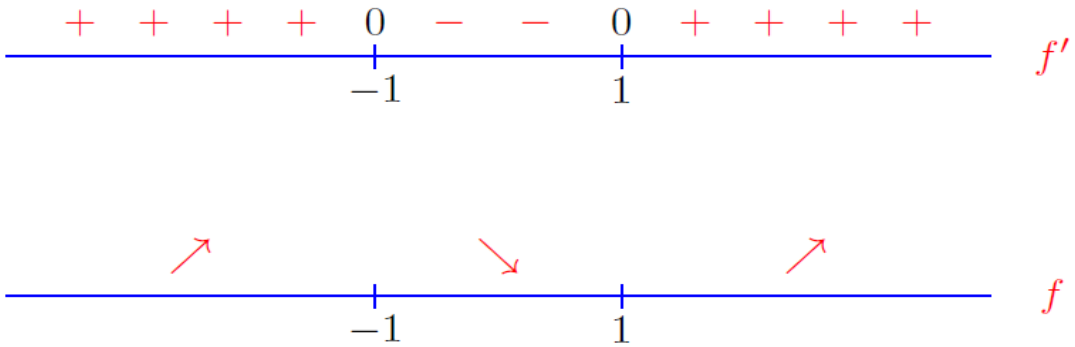
f increasing on open intervals

- $(-\infty, -1)$
- $(1, \infty)$

f decreasing on open interval

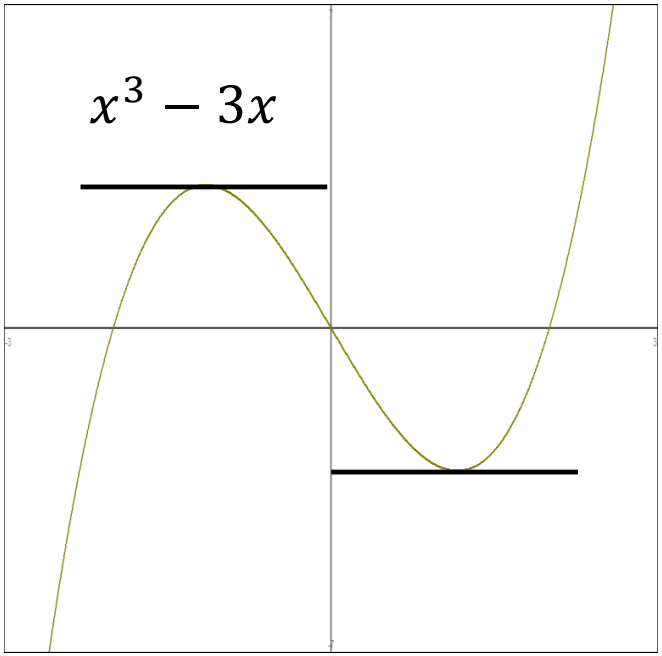
- $(-1, 1)$

Sign diagram:



Stationary Points

If $f'(c) = 0$, then c is a stationary point of f

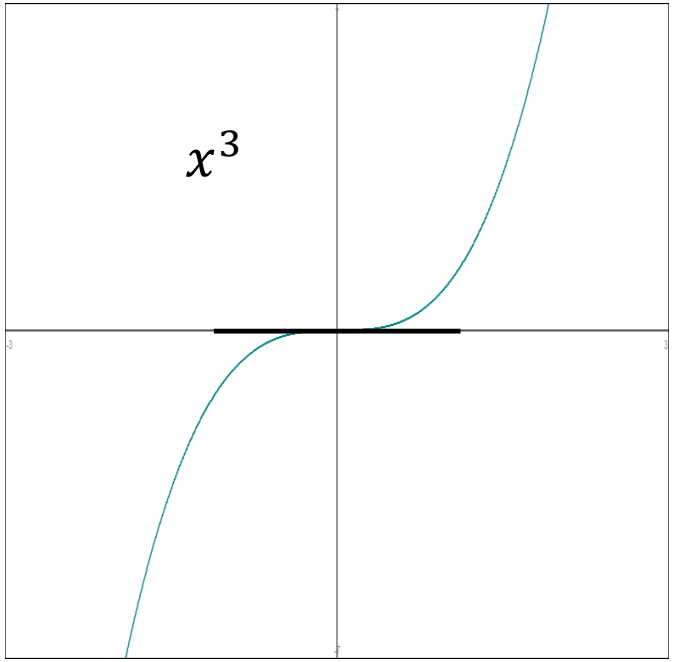


$$f'(x) = 3x^2 - 3$$

$+$ $-$ $+$

-1 1

Change of sign (twice)



$$f'(x) = 3x^2$$

$+$ $+$

0

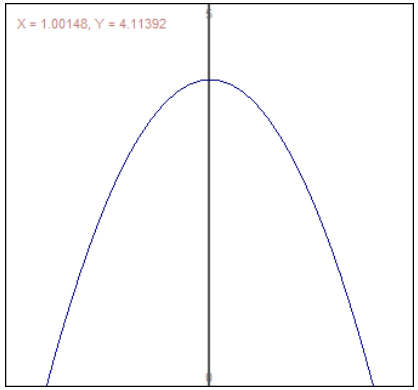
No change of sign

Extreme Points - Maximum Profit and Minimum Cost

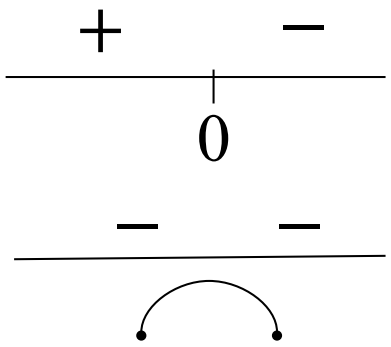
- If $f'(x)$ moves **from positive to negative** at a point c (i.e., $f''(c) < 0$),

  c is a (local) maximum point for f

  (If f measures profits, then c maximizes these profits)



- $f(x) = 4 - x^2$
- $f'(x) = -2x$
- $f''(x) = -2$

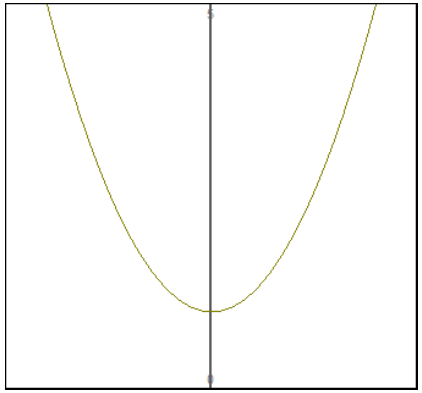


Extreme Points - cont'd

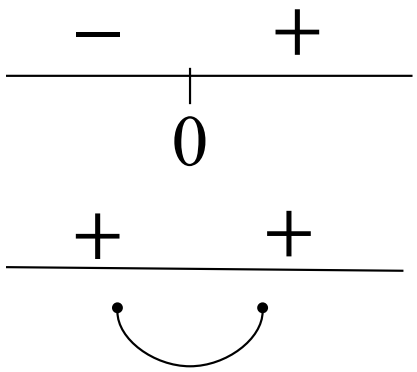
- If $f'(x)$ moves **from negative to positive** at a point c(i.e., $f''(c) > 0$),

  c is a (local) minimum point for f

  (If f measures costs, then c minimizes these costs)



- $f(x) = x^2 + 1$
- $f'(x) = 2x$
- $f''(x) = 2$



Convexity and Concavity

Consider a function f defined on an open interval I.
If, for all $x, y \in I$ and all $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y) \tag{1}$$

then f is called concave (on I)

- If $f''(x) \leq 0$, for all $x \in I$, then f is concave
- If the inequality in (1) is strict, then f is strictly concave

If, for all $x, y \in I$ and all $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \tag{2}$$

then f is called convex (on I)

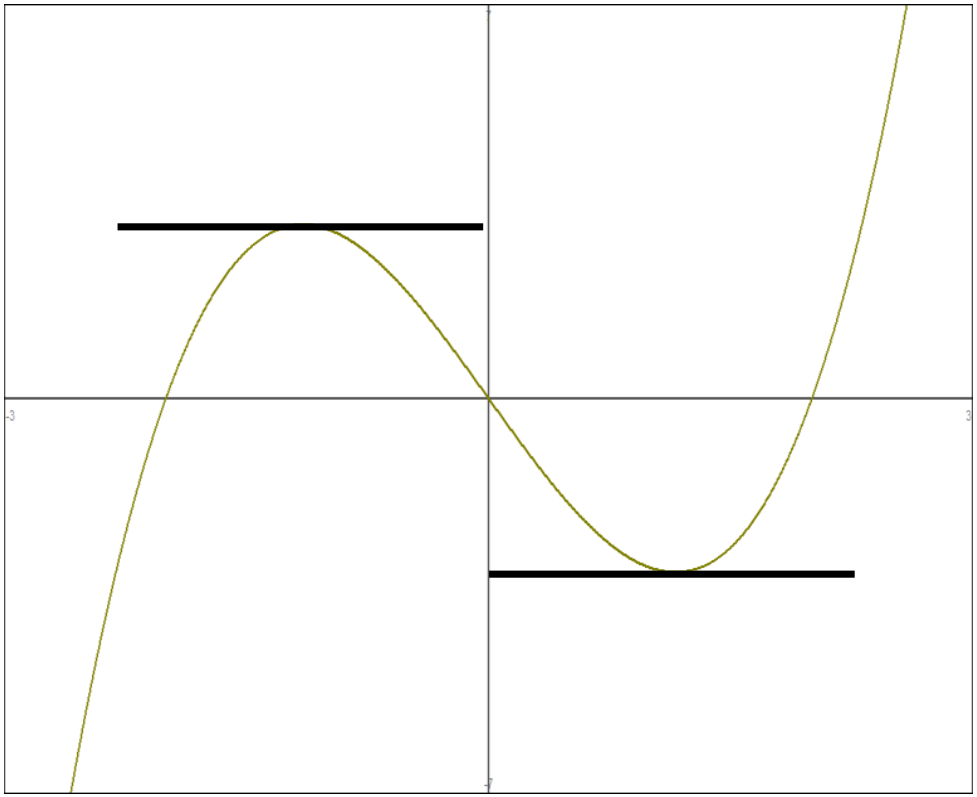
- If $f''(x) \geq 0$, for all $x \in I$, then f is convex
- If the inequality in (2) is strict, then f is strictly convex

Convexity and Concavity - Applied

Concavity/Convexity is the key to understanding the efficiency of a process.

- Concave Function ($f'' < 0$):
  - This represents diminishing marginal returns
  - If this is a production function, each additional employee adds less output than the last did
- Convex Function ($f'' > 0$):
  - This represents increasing marginal returns
  - Each additional input is more productive than the last was
- Inflection Point ($f'' = 0$):
  - Switching point from convex to concave or from concave to convex

Convexity and Concavity - cont'd



If, for all $x, y \in I$ and all $\lambda \in [0, 1]$,

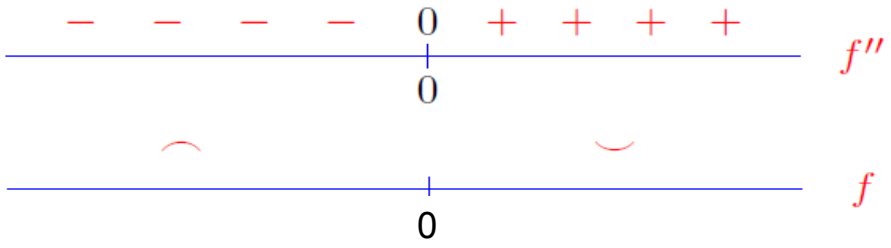
$$f(\lambda x+(1-\lambda)y) \geq \lambda f(x)+(1-\lambda)f(y)$$

then f is called concave (on I)

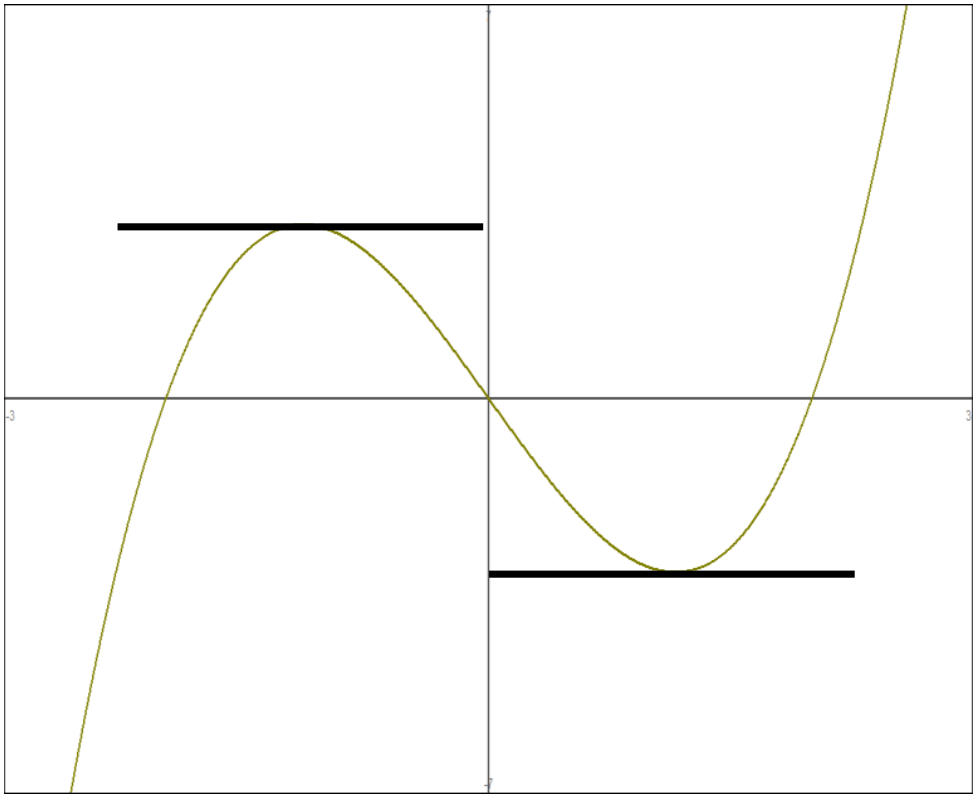
If, for all $x, y \in I$ and all $\lambda \in [0, 1]$,

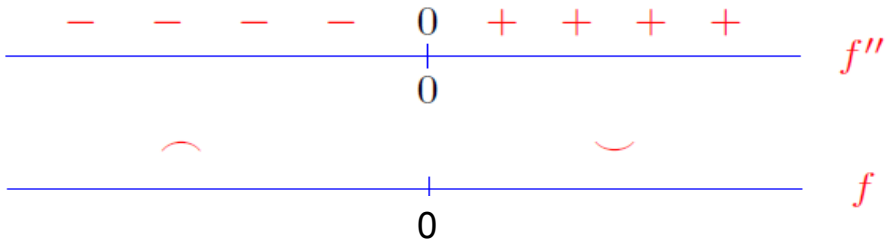
$$f(\lambda x+(1-\lambda)y) \leq \lambda f(x)+(1-\lambda)f(y)$$

then f is called convex (on I)



Inflection Points





What happens with f at $x = 0$?

- $f(x) = x^3 - 3x$ changes from concave to convex
- $f'(x) = 3x^2 - 3$ changes from decreasing to increasing
- $f''(x) = 6x$ changes its sign

In general, if g'' changes sign at $x = c$, then the point $(c, g(c))$ is called an inflection point. At such a point, g either changes from convex to concave or from concave to convex.

Types of Extreme Values

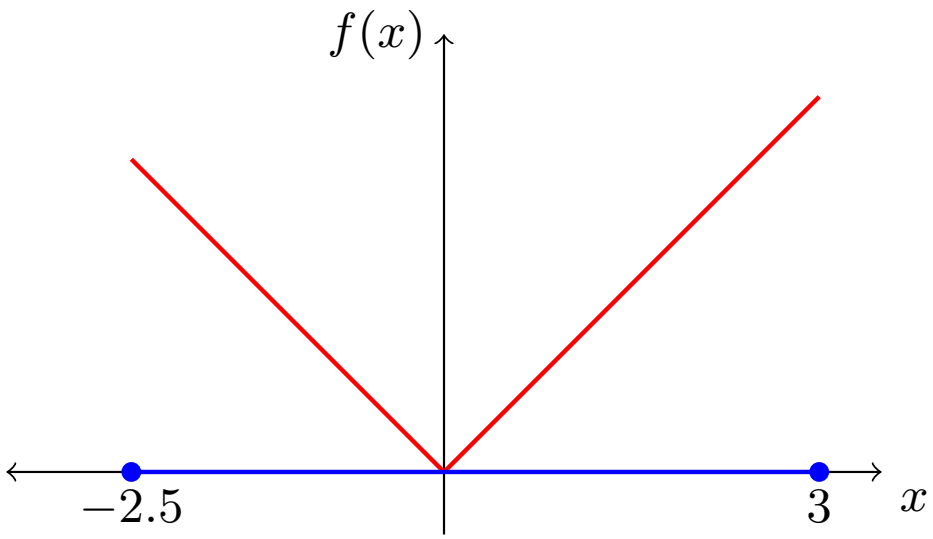
THREE TYPES OF EXTREME VALUES

If a function f has an extreme value at c, then one of the following three holds

(1) c is a stationary point of f, i.e., $f'(c) = 0$, or

(2) c is a point where f is not differentiable, or

(3) c is an endpoint of the domain of the function



$$f(x) = \begin{cases} -x, & \text{if } -2.5 \leq x \leq 0 \\ x, & \text{if } \quad 0 \leq x \leq 3 \end{cases}$$

- has maximum at $x = -2.5$
- has maximum at $x = 3$
- has minimum at $x = 0$

Recipe for Drawing Functions

1. Find the domain of f
2. Find the zeroes (roots) of f
3. Determine f'
   a) Compute the zeroes of f' (stationary points of f)
   b) Compute the sign-survey
4. Determine f''
   a) Compute the zeroes of f'' (stationary points of f')
   b) Compute the sign-survey

Example

Consider the function $y = x^2 e^{-x}$

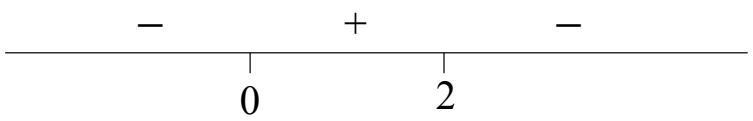
1. Domain: $\mathbb{R}$

2. Zeroes: $x = 0$

3. 1st order derivative: $y' = 2xe^{-x} + x^2(-1)e^{-x} = (2x - x^2)e^{-x}$

   a) Zeroes: $y' = 0 \quad \Leftrightarrow \quad (2x - x^2)e^{-x} = 0 \quad \Leftrightarrow \quad x = 0$, or, $x = 2$

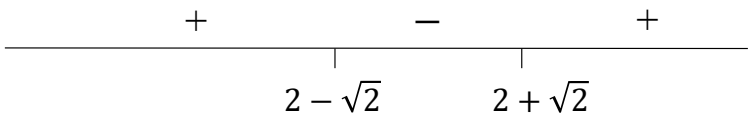
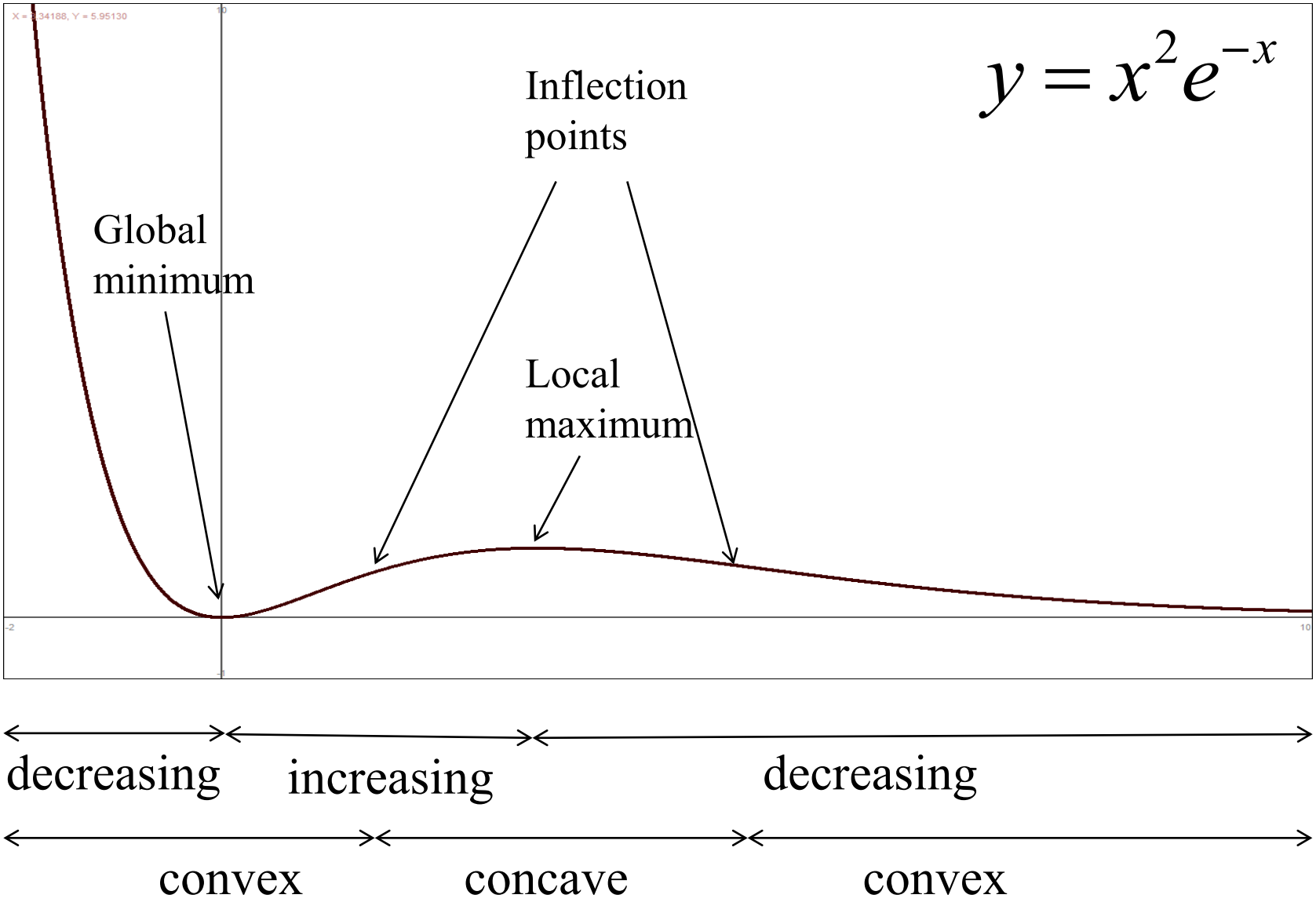
   b) Sign survey:



4. 2nd order derivative $y'' = (2 - 2x)e^{-x} + (2x - 2x^2)(-1)e^{-x} = (2 - 4x + x^2)e^{-x}$

   a) Zeroes: $(2 - 4x + x^2)e^{-x} = 0 \quad \Leftrightarrow \quad x = 2 - \sqrt{2}$, or, $x = 2 + \sqrt{2}$

   b) Sign survey:



Example - cont'd